TRAUMA · NEET-SS / FET / INI-SS

Fracture healing and fixation principles

1. Bone healing

Primary (direct) bone healing

- Occurs with **absolute stability** (anatomical reduction and **interfragmentary compression**, strain under about 2%).
- **Cutting cones**: osteoclasts at the tip cross the fracture, osteoblasts follow laying down lamellar bone (**Haversian remodelling**). **Contact healing** (gap under about 0.01 mm) and **gap healing** (gap under about 0.8-1 mm: woven bone first, then remodelling). **No callus.**

Secondary (indirect) bone healing

- Occurs with **relative stability** (casts, nails, bridging plates, external fixators; strain about 2-10%). **Callus formation** via endochondral (and intramembranous) ossification.
- **Stages:**
  - **Haematoma and inflammation** (hours to days): platelets, macrophages; cytokines **IL-1, IL-6, TNF-alpha, PDGF, TGF-beta**; mesenchymal stem cell recruitment.
  - **Soft callus (repair)** (about 2-3 weeks): chondrocytes produce **type II collagen** cartilage; periosteal intramembranous bone.
  - **Hard callus** (about 3-4 months): endochondral ossification, woven bone (**type I collagen**); **type X collagen** in hypertrophic chondrocytes during calcification.
  - **Remodelling** (months to years): woven to lamellar bone along stress lines (**Wolff's law**); **piezoelectric** effect (compression side negative: bone formation; tension side positive: resorption).
- **Perren's strain theory:** strain = change in gap length / original gap length. **Under 2%: primary bone; 2-10%: bone (secondary); 10-100%: cartilage/fibrous tissue; over 100%: granulation tissue/non-union.** A small gap with too much motion has very high strain (non-union); comminution distributes strain.

- **The "diamond concept"** (Giannoudis): osteogenic cells, osteoconductive scaffold, growth factors (osteoinduction), **mechanical stability**, plus vascularity and host factors.

Growth factors and biologics

- **BMPs** (TGF-beta superfamily; bind serine-threonine kinase receptors, signal via **SMAD**). **rhBMP-2: FDA approved for open tibial fractures treated with IM nails** (BESTT trial) and anterior lumbar interbody fusion; **rhBMP-7 (OP-1)** humanitarian exemption for recalcitrant non-unions (withdrawn). Complications: heterotopic bone, swelling (cervical spine), possible cancer concerns.
- **Bone graft properties: osteogenic** (living cells: **autograft** only, and bone marrow aspirate), **osteoinductive** (BMPs; autograft, demineralised bone matrix), **osteoconductive** (scaffold: autograft, allograft, calcium phosphate/hydroxyapatite, tricalcium phosphate, calcium sulfate).
- **Autograft (iliac crest)** is the gold standard (all three properties). **Cancellous** autograft: rapid incorporation (osteogenic, creeping substitution); **cortical**: structural strength, slower (resorption first: weaker at 6 weeks-6 months).
- **RIA (reamer-irrigator-aspirator)** femoral graft: large volume, high osteogenic content.
- **Vascularised fibula:** for defects over about 6 cm, hypertrophies with loading.
- **Masquelet (induced membrane) technique** for segmental defects: **stage 1** debridement + **PMMA cement spacer** (induces a vascular membrane rich in growth factors, VEGF, BMP-2) ; **stage 2** (about 4-8 weeks) remove spacer, fill the membrane with cancellous autograft (± RIA).

Factors impairing healing

- **Smoking** (nicotine vasoconstriction, inhibits osteoblasts; **increases non-union** substantially), **diabetes**, **NSAIDs** (COX-2 inhibition; short courses probably safe; avoid in high-risk non-union cases), corticosteroids, **malnutrition** (albumin under 3.5 g/dL, lymphocytes under 1,500), vitamin D deficiency, infection, **excessive motion or excessive rigidity with a gap**, **open fractures with periosteal stripping**, high-energy injuries, **compromised blood supply** (scaphoid, talar neck, femoral neck, fifth metatarsal zone 2, tibial distal third), anaemia, hypothyroidism, quinolones (animal data), bisphosphonates (delay remodelling, do not stop union).

- **Pulsed electromagnetic fields, LIPUS** (low-intensity pulsed ultrasound: **TRUST trial** found no benefit in tibial fractures).

2. Non-union

- **Definition:** failure to heal with **no progression of healing for 3 consecutive months** (FDA: 9 months after injury). **Delayed union:** healing slower than expected.
- **Weber-Cech classification:**

Type	Biology	Appearance	Cause	Treatment
Hypertrophic ("elephant foot", "horse hoof")	**Biology good** (vascular)	Abundant callus that fails to bridge	**Instability (too much motion)**	**Improve stability**: exchange (reamed) nailing, compression plating; **no graft needed**
Oligotrophic	Moderate	Little callus	Inadequate reduction/distraction	Stability + biology
Atrophic	**Poor biology** (avascular)	No callus, rounded/pencilled ends	Poor blood supply, infection, smoking, stripping	**Debride to bleeding bone + bone graft/biologics + stable fixation**
Pseudarthrosis (synovial)		Fluid-filled false joint	Persistent motion	Excise + graft + fixation

- **Infected non-union:** debridement, deep cultures, dead space management, stabilisation, local and systemic antibiotics, staged reconstruction (Masquelet, Ilizarov bone transport).
- **Work-up of any non-union: exclude infection** (CRP, ESR, WBC, deep tissue samples, nuclear imaging), **metabolic/endocrine screen** (vitamin D, calcium, thyroid, PTH, testosterone, glucose; Brinker's endocrine abnormalities found in a majority of unexplained non-unions), CT to confirm.

- **Exchange nailing:** for **aseptic diaphyseal non-union** (especially femur): ream at least 1-2 mm more and insert a larger nail. Less successful in the tibia with bone loss and in metaphyseal non-unions.
- **Distraction osteogenesis / bone transport (Ilizarov): latency 5-7 days**, distraction **1 mm/day in 4 increments (0.25 mm every 6 hours)**, then consolidation (about 1 month per cm in adults; **healing index**). Too fast: fibrous tissue/poor regenerate; too slow: premature consolidation. **Corticotomy** preserves the medullary blood supply and periosteum (low-energy). Regenerate forms by **intramembranous ossification**.

3. Biomechanics of fixation

Absolute vs relative stability

	Absolute stability	**Relative stability**
Healing	**Primary (no callus)**	**Secondary (callus)**
Indicated for	**Articular fractures**, simple (two-part) diaphyseal fractures of the forearm, simple metaphyseal fractures	**Comminuted diaphyseal and metaphyseal fractures**
Methods	**Lag screws, compression plating, tension band**	**IM nails, bridging plates, external fixators, casts**
Requirement	**Anatomical reduction**	Restore length, alignment and rotation; preserve biology

Screws

- **Lag technique:** the thread engages only the far cortex (**overdrilled "gliding hole" in the near cortex**); compresses fragments; place **perpendicular to the fracture** (best compression) or bisecting the angle between perpendicular to the fracture and to the shaft.
- **Pull-out strength** depends on the **major (outer) diameter**, thread depth and pitch, and bone density. **Bending/fatigue strength** depends on the **core (root) diameter** (strength proportional to the **radius to the fourth power** for bending stiffness, third power for strength).
- **Position screws** (hold fragments without compression), **locking screws** (threaded heads lock into plates: fixed-angle construct), **cannulated screws**

(smaller core: weaker).

Plates (function, not design)

Function	Description	Example
Compression	Dynamic compression holes (eccentric screw) or tensioning device compress a transverse fracture	Forearm shaft (LC-DCP)
Neutralisation (protection)	Protects a lag-screw fixation from bending/torsion	Lag screw + one-third tubular plate in a Weber B fibula
Buttress (antiglide)	Resists shear/axial load at a metaphysis	Tibial plateau, volar Barton's, posterior antiglide fibula
Tension band	On the **tension (convex) side** converts tension into compression	Olecranon wiring, patella, femoral plating laterally
Bridging	Spans comminution; relative stability	Comminuted shaft (MIPO)

- **LC-DCP** (limited contact): undercuts reduce periosteal contact, preserve blood supply, even stiffness.
- **Locking plates:** fixed-angle; do not depend on bone-plate friction; **advantageous in osteoporotic bone, short metaphyseal segments, periprosthetic fractures**; act as an "internal external fixator". **Too stiff** (short working length, all holes filled near the fracture) can **suppress callus** (**far cortical locking / dynamic locking screws** add flexibility). Bridging rule: plate span about **2-3 times the fracture length**, **screw density under 0.4-0.5**, leave 2-3 holes empty over the fracture.
- **Working length:** distance between the closest screws on either side of the fracture; longer working length = more flexible construct.

Intramedullary nails

- **Load-sharing** devices (vs load-bearing plates). **Bending stiffness proportional to the fourth power of the radius** (a slightly larger nail is much stiffer); slotted nails have reduced torsional stiffness.

- **Locking: static** (screws both ends: controls length and rotation), **dynamic** (allows compression; for stable patterns).
- **Reaming:** increases nail size and union rates; produces **autograft** at the fracture; temporarily reduces endosteal blood flow (periosteal flow increases compensatorily); fat intravasation.
- **Poller (blocking) screws:** narrow the medullary canal in metaphyseal fractures to prevent angulation (placed on the **concave side of the deformity**).

External fixation

- **Stiffness increases with: larger pin diameter** (most important; stiffness proportional to radius^4), **pins closer to the fracture** and further apart on each fragment (**near-far spread**), **bars closer to the bone**, **more bars (double-stacked)**, **multiplanar** configuration, more pins.
- **Pin-site infection** is the commonest complication; **hydroxyapatite-coated pins** improve fixation in cancellous/osteoporotic bone.
- **Circular (Ilizarov) frames:** tensioned thin wires (90-130 kg tension); allow axial micromotion ("trampoline effect") that promotes healing; deformity correction and bone transport.

Material properties (implants vs bone)

- **Elastic (Young's) modulus**, from highest to lowest: **ceramic (alumina) > cobalt-chromium > stainless steel > titanium alloy > cortical bone > PMMA > polyethylene > cancellous bone > cartilage**.
- **Titanium:** modulus closest to bone (less stress shielding), excellent biocompatibility, **notch-sensitive**, poor wear resistance as a bearing.
- **Stress shielding:** stiff implants carry load, bone underneath resorbs (refracture after plate removal: protect for about 6 weeks).
- **Fatigue failure:** implants break if the fracture does not heal ("**race between healing and implant failure**").

4. Principles summary (AO)

1. **Fracture reduction and fixation to restore anatomical relationships.**
2. **Stability by fixation or splintage** appropriate to the fracture and injury (absolute vs relative).

3. **Preservation of the blood supply** to soft tissues and bone by gentle handling and careful reduction (MIPO, indirect reduction).
4. **Early and safe mobilisation** of the part and the patient.

Exam pearl: **Absolute stability = primary healing, no callus, anatomical reduction (articular, simple forearm shaft). Relative stability = callus (comminuted diaphyses: nails, bridging plates).** Perren: strain under 2% bone primary; over 10% fibrous.

Exam pearl: **Hypertrophic non-union = needs stability (no graft); atrophic = needs biology (graft) + stability. rhBMP-2 is FDA approved for open tibial fractures with IM nails.** Ilizarov: latency 5-7 days, 1 mm/day in 4 steps.

Exam pearl: **External fixator stiffness: pin diameter (most important), pins near the fracture and far apart, bar close to bone, double bar. Modulus: ceramic > CoCr > steel > titanium > cortical bone.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.